

14.2.2.3. TLS Root CA Certificate (TLSRCAC)

A TLS Root CA Certificate (TLSRCAC) is a special type of digital certificate that plays a fundamental role in the Public Key Infrastructure (PKI). It is the topmost certificate in the hierarchical structure of the certificate chain and holds the highest level of trust. TLS clients use TLSRCACs to authenticate a server during the handshake process, using the "Chain of Trust" described in [RFC 4158](#).

14.2.2.4. TLS Client Certificate

A TLS Client Certificate is a type of digital certificate used to authenticate clients during the TLS handshake. Use of client certificates in TLS is covered in [RFC 8446](#) (or later).

14.2.2.5. Certificate Signing Request (CSR)

A Certificate Signing Request (CSR) is used by an entity to request a CA to issue a digital certificate to that entity. The CSR contains information about the entity's public key and MAY contain other relevant information. All CSRs in this chapter SHALL be created following the format and procedure in [PKCS #10](#).

14.2.3. TLS Endpoint

A TLS Endpoint is a specific network address or hostname to which communication using TLS can be established as defined in [RFC 8446](#) (or later), and thus represents a potential connection from the Node to another device. A TLS Endpoint can be either a TLS Client, which is the TLS Endpoint initiating the connection, or a TLS Server, which is the TLS Endpoint serving the requests from the TLS Client. In this chapter the Node is presumed to be the TLS Client, and the TLS Server is some other TLS Endpoint that the Node wishes to connect to.

In the context of this chapter, a TLS Endpoint is used to refer to the combination of a TLS Hostname and a TLS Port Number, along with certain other details, representing a TLS Server.

TLS Endpoints are managed by the [TLS Client Management Cluster](#).

14.2.3.1. TLS Hostname

A TLS Hostname SHALL be a domain name or IP Address. For cloud based TLS servers this MAY be a public resolvable DNS name. For local TLS servers on the same broadcast domain or locally reachable IP Addresses, this MAY be a multicast DNS based .local name, or other resolvable private name.

14.2.3.2. TLS Port Number

A TLS Port Number is the network port designated for TLS connections. If a Port Number is omitted, the value of 443 SHALL be assumed.

14.2.4. TLS Connection

A TLS Connection is defined as a connection between two TLS Endpoints, one of which is the TLS client side and the other as the TLS server side. Establishment and usage of all TLS Connections uses the standard procedure defined in [RFC 8446](#) (or later). A TLS Client Connection is defined as a TLS Connection where the Node is the TLS client side.